

Siddharth Nair

SOFTWARE ENGINEER I · JUNIOR SOFTWARE ENGINEER

346-777-6360 | nair.sidwork@gmail.com | NairSiddharth | siddharthnair01

Summary

Motivated software engineer eager to tackle challenging problems across data engineering, machine learning, and full-stack application development. Proficient in Python, SQL, and Java, with hands-on experience creating scalable data pipelines, predictive models, user-facing tools, and prompt engineering solutions to enable advanced AI-driven workflows. Looking to contribute to innovative teams where I can apply my technical skills and collaborative mindset to deliver impactful, real-world solutions.

Education

Texas A&M University

COMPUTER ENGINEERING

College Station, TX

Aug. 2019 - Dec. 2024

- Honors: National Merit Scholarship, Presidents Endowed Scholarship
- Relevant Coursework: CSCE 452 - Robotics, CSCE 420 - Artificial Intelligence, CSCE 411 - Analysis of Algorithms, CSCE 313 - Introduction to Computer Systems, CSCE 315 - Programming Studio, CSCE 464 - Wireless & Mobile Systems

Work Experience

JP Morgan Chase

Houston, TX

Plano, TX

SOFTWARE ENGINEERING INTERN

Jun. -- Aug. 2022--2025

- **2025:** Delivered multiple large-scale initiatives impacting **1,000+ engineers** across Chase Digital and indirectly **85M+ customers**:
 - Automated export and normalization of 100+ Akamai CDN properties into Terraform modules with dynamic `.tfvars` creation, enabling faster, more reliable configuration of all public-facing Chase websites.
 - Developed a multimodal ingestion pipeline for an Agentic RAG application, enhancing search across architectural documentation with role-specific prompt engineering.
 - Enhanced search relevance by combining lexical (BM25) and semantic (vector) search through Reciprocal Rank Fusion (RRF).
 - Implemented a Redis cache POC for frequently queried vector embeddings, reducing time taken for FAQ retrieval by minimizing queries of the full vector DB.
- **2024:** Deployed tooling and forecasting models to accelerate backend operations, reducing production bug resolution time by **50%**:
 - Designed web-based automation for parameterized batch jobs, streamlining environment migrations and bug resolution.
 - Engineered a time-series forecasting model with **5.38% SMAPE**, leveraging historical data and macroeconomic indicators to improve accuracy.
- **2023:** Developed a full-stack analytics dashboard for the Fundamental Review of the Trading Book (FRTB) team, enabling real-time evaluation of trader performance against Market Risk benchmarks.
 - Introduced dynamic filtering and historical drill-down capabilities to enhance pattern detection and compliance analysis.
- **2022:** Built an internal alternative to Comet.com for model monitoring and control:
 - Developed an interactive dashboard enabling both real-time monitoring and direct control of ML training jobs and deployment status, with live logging and visual analytics.

Texas A&M University Department of Recreational Sports

College Station, TX

INTRAMURAL FLAG FOOTBALL SUPERVISOR

Jan. 2023 -- Dec. 2024

- Contributed to managing intramural flag football operations for **14,000+ participants** per semester, managing referee teams, and supervising tournaments while officiating games.
 - Coached referees through targeted training and one-on-one mentoring, with several advancing to supervisory roles.
 - Recognized for conflict management and rules knowledge; selected to officiate special events and regional tournaments.

AI4ALL

College Station, TX

STUDENT LEADER

Jan. 2023 -- Aug. 2024

- Facilitated Apply AI program events and guided 4 student groups per term through project milestones, collaborating with industry partners to ensure alignment with real-world applications.
- Taught core algorithm concepts and model training techniques; coached students on showcasing project impact effectively.

Lumen Technologies

Houston, TX

SOFTWARE ENGINEERING INTERN

Jun. 2021 -- Aug. 2021

- Engineered a Python program to generate daily flat files from database inputs for logging and reporting; implemented monthly database resets with shell scripts to prevent data over-accumulation and maintain system performance.
- Refined SQL scripts to accurately produce billing reports for **10,000+ customers**.

Skills

Languages Python, SQL (Oracle, Postgres, MySQL), Java, JavaScript

DevOps/Infrastructure Terraform, Jules, Spinnaker, Bash, Git | AWS Athena, Aurora, Elasticache for Redis, OpenSearch, S3

Backend Azure OpenAI, LangChain, REST APIs, Pandas, Pydantic, Springboot, FastAPI, AWS Sagemaker

Frontend React, HTML5, Voila, Plotly, ipywidgets